

Unconcentrated markets, persistent ties: incumbency in Türkiye’s public IT procurement

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Abstract

Market-level concentration indices are the standard first screen for competition problems, but they cannot show how often a public buyer returns to the same supplier. We study this with 9,991 information-technology (IT) contracts awarded on Türkiye’s e-procurement platform EKAP between 2010 and 2026. Supply is unconcentrated in every product market. Yet 45% of contracts awarded by a buyer that had bought in the same market before go to an incumbent supplier. That is 1.7 times the rate under the most conservative permutation null, which holds each buyer’s demand and each firm’s wins fixed within market, year, and the provinces of both parties. About a third of these contracts renew recurring services, and they carry most of the excess. To ask how much of this persistence reflects switching costs, we model who wins among firms active before each tender. We compare the incumbent advantage with a placebo for future suppliers and benchmark both against simulations of pure state dependence and pure heterogeneity. Neither pure mechanism fits. Persistence is dominated by stable buyer–firm heterogeneity, which may reflect fit, relationships, or favouritism; a state-dependence component consistent with switching costs appears only in renewals. Incumbent wins coincide with more single bidding and, in a sample of 143 tenders, with smaller discounts on the estimated cost, although that difference is not statistically significant. Buyer-level persistence, benchmarked against null models, is a useful screen alongside market concentration.

Keywords: Public procurement, Incumbency, Switching costs, State dependence, Market concentration, Türkiye

JEL Classification: H57 , L14 , L86 , D22

1 Introduction

Competition authorities and procurement regulators usually begin with market structure. The Herfindahl–Hirschman index (HHI) and concentration ratios summarize how supply is shared among the firms in a relevant market, and fixed thresholds turn them into a first judgement about competitive risk (Hirschman, 1964; Rhoades, 1993; U.S. Department of Justice and Federal Trade Commission, 1992, 2023). These measures cannot show whether individual buyers keep returning to the same firm. In public IT, where installed systems create switching costs, such persistence is the natural worry: once a buyer has a system in place, competition for its later purchases can be weak even when the market as a whole is competitive (Borenstein, MacKie-Mason, & Netz, 2000; Farrell & Klemperer, 2007; Farrell & Shapiro, 1988; Greenstein, 1993; Klemperer, 1987; Shapiro, 1995).

Persistence, however, is not the same as lock-in. A buyer may return to a supplier because switching is costly (state dependence), or because that supplier simply suits its needs, location, or systems, or because the two have a relationship that the buyer values or that favours the supplier (persistent heterogeneity). Separating these explanations is a classic problem (Dubé, Hitsch, & Rossi, 2010; Heckman, 1981; Heckman & Borjas, 1980; Keane, 1997), and it matters for policy. Switching costs call for remedies that lower them. Heterogeneity may be benign, as with specialization and good matches, or it may reflect favouritism, which calls for scrutiny of discretion.

This paper asks how much buyer–supplier persistence there is in Türkiye’s public IT procurement, and how much of it looks like state dependence. We reconstruct 9,991 awarded IT contracts from the national e-procurement platform EKAP (2010–2026), linking 2,814 firms to 2,890 buyers. The analysis proceeds in four steps.

- First, supply is unconcentrated. Count-based HHIs are below 1,000 in every product market, so a concentration screen would raise no flag.
- Second, persistence is large. Of the contracts awarded by a buyer that had bought in the same market before, 45.2% went to an incumbent. That is 1.7 times the rate under the most conservative permutation null, which keeps each buyer’s demand and each firm’s wins fixed within product market, year, and the provinces of both parties.
- Third, about a third of these contracts renew a recurring service, and they carry most of the excess.
- Fourth, we model which firm wins among those active in the market before each tender. We compare the advantage of past suppliers with that of firms that will supply the buyer only in the future, and benchmark both against simulated worlds with pure state dependence and pure heterogeneity. Neither world fits. Persistence is dominated by stable buyer–firm heterogeneity. A state-dependence component consistent with switching costs appears in renewals but not in new needs.

Persistence is highest in health information systems and under the negotiated procedure without publication, and incumbent-won tenders attract single bids far more often. In a sample of tenders for which we collected estimated costs, incumbents grant smaller discounts, but the difference is imprecise.

We contribute in three ways. First, recent work has combined concentration measures with network analysis of procurement markets, showing that concentration indices and centrality can point in different directions (I. Fountoukidis, Daffi, Antoniou, & Varsakelis, 2026; I.G. Fountoukidis, Antoniou, & Varsakelis, 2023; Pliatsidis, 2024). We shift the focus from the structure of the network to the persistence of its ties, benchmark that persistence against null models, and decompose it with a lead-lag placebo and simulation benchmarks. This shows that persistence is mostly heterogeneity rather than lock-in. Second, we separate renewals of recurring services from new needs, which isolates where switching costs plausibly operate. Third, we provide evidence on Turkish public IT procurement, a market not studied before, adding to work on political connections and construction (Gürakar, 2016; Gürakar & Bircan, 2019; Kutlay & Yıldırım, 2026). We also document a value-based audit of keyword-defined EKAP data.

2 Background

2.1 Switching costs, heterogeneity, and incumbency

The HHI is informative only relative to a relevant market; pooled across markets that are not substitutes, it is mechanically diluted (Rhoades, 1993). In procurement the relevant choice set can be narrower still. The switching-cost and aftermarket literatures show how a supplier with modest market share can hold substantial power over its installed base (Borenstein et al., 2000; Farrell & Shapiro, 1988; Klemperer, 1987; Shapiro, 1995; Shapiro & Varian, 1999), and the European Commission has warned of such lock-in in public IT (European Commission, 2013). Observed inertia can reflect such costs or persistent heterogeneity in preferences and fit. The two are distinguished by how past and future choices relate, as in lead-lag tests (Dubé et al., 2010; Heckman, 1981; Heckman & Borjas, 1980; Keane, 1997; Shcherbakov, 2016).

In procurement, incumbents can hold cost or information advantages that shape entry and bidding (De Silva, Dunne, & Kosmopoulou, 2003; Hyttinen, Lundberg, & Toivanen, 2018; Jofre-Bonet & Pesendorfer, 2003; Li & Zheng, 2009; Saini, 2012), and limited competition in government purchasing is common (Kang & Miller, 2022). Buyer discretion and the people who exercise it matter (Best, Hjort, & Szakonyi, 2023). More discretion raises the share of repeat winners without necessarily worsening performance (Coviello, Guglielmo, & Spagnolo, 2018), but it can also be used to favour connected firms, at a measurable cost (Baltrunaite, Giorgiantonio, Mocetti, & Orlando, 2021; Bařanek & Titl, 2024; Decarolis, Fisman, Pinotti, & Vannutelli, 2025; Palguta & Pertold, 2017; Szucs, 2024). Relationships that develop with a buyer's tenure can sustain repeat awards (Coviello & Gagliarducci, 2017). Relational contracts can make loyalty to proven suppliers valuable (Board, 2011; Calzolari & Spagnolo, 2009). Past performance legitimately informs supplier choice, although it may deter entry (Butler, Carbone, Conzo, & Spagnolo, 2020; Decarolis, Spagnolo, & Pacini, 2016), and negotiation can be efficient for complex purchases (Bajari, McMillan, & Tadelis, 2009). Persistence is therefore not by itself evidence of harm; its consequences depend on prices and quality (Bandiera, Prat, & Valletti, 2009).

2.2 Procurement as a network

Contract-award records are naturally two-mode. Network studies of procurement have examined corruption risk and favouritism (Fazekas & Wachs, 2020; Wachs, Fazekas, & Kertész, 2021), collusion in co-bidding networks (Reeves-Latour & Morselli, 2017; Wachs & Kertész, 2019), and the structure of buyer–supplier transactions (Popa, 2019). Integrity indicators built from award data include single bidding and the concentration of a buyer’s spending on one supplier (Fazekas & Kocsis, 2020; Fazekas, Tóth, Abdou, & Al-Shaibani, 2024; Fazekas, Tóth, & King, 2016). I.G. Fountoukidis et al. (2023) show for the European pacemaker market that concentration-type indices can signal competition while centrality reveals dominant suppliers, and Pliatsidis (2024) measures concentration in Greek procurement with network methods. Our object is the buyer–supplier tie rather than node centrality, and our contribution is to benchmark its persistence and decompose it.

2.3 The Turkish context

Turkish public procurement is governed by Law No. 4734 (2002) and administered by the Public Procurement Authority (KİK), whose platform EKAP publishes notices and awards (Alsaç, 2017); see Buğra and Savaşkan (2014) and Gürakar (2016) on its political economy. Two grounds for negotiation matter here. Under Article 21(b), which covers urgency after unforeseen events, and Article 21(f), which covers purchases of goods and services below an annually indexed limit, the buyer need not publish a notice and instead invites at least three firms of its choosing. Law No. 7144 (2018) extended 21(b) to certain works (Alyanak, 2022). Politically connected firms have been found to win more often under less competitive procedures (Gürakar, 2016; Gürakar & Bircan, 2019), and market capture intensified during the 2010s (Kutlay & Yıldırım, 2026). In the health sector, Decree-Law No. 694 of August 2017 abolished the Public Hospitals Authority and its provincial hospital unions and returned hospital procurement to the Ministry of Health and to individual hospitals.

3 Data

3.1 Collection and scope

We collected award records of IT-related tenders from EKAP by keyword search (13,418 records with an identifiable winner). EKAP search matches substrings of titles, so it admits tenders in which an IT term appears only incidentally. For example, the keyword *ERP* matches *tuz serpici* (salt spreader), and *technical support* matches hospital staffing. An earlier count-based cleaning left 13,024 contracts, but a value-based audit showed that errors were concentrated among large contracts. We classified the 150 largest contracts by hand, applied ordered title rules to the rest, and inspected samples from every rule. Of the 13,024 contracts, 1,627 (12.5%, holding 17.7% of nominal value) are not IT, such as housing construction, vehicles, and cleaning services. A further 1,406 are ambiguous, such as data-entry staffing or building works with security cameras. The main sample contains the 9,981 unambiguous IT contracts and 10 contracts for IT-enabled call-centre services, one of which is the largest in the data

Table 1 Main sample (IT and IT-enabled services, EKAP 2010–2026).

Awarded contracts	9,991
Firms (canonicalized) / buyers (pooled labels split)	2,814 / 2,890
Buyer–firm pairs; with two or more contracts	7,219; 1,377 (19.1%)
Repeat-eligible contracts (buyer bought in the market before)	4,913
Total value, 2025 TRY (nominal TRY)	89.3 bn (33.3 bn)
Largest contract, share of real value	8.4%
Procedure: open / Art. 21(b) / Art. 21(f) / other	59.0 / 10.5 / 29.6 / 0.9%
Tender dates	Oct 2010 – Mar 2026

(the 2025 national hospital appointment call centre); value-based results are reported with and without it. EKAP lists one winner per tender, so multi-lot tenders appear with a single winner (about 116 titles suggest lots); we found no framework-agreement call-offs.

3.2 Entities, markets, and values

Firm names were canonicalized by normalizing characters and stripping legal-form suffixes, with the merge groups of the 300 largest firms reviewed by hand. This reduces 4,788 recorded names to 4,389 firms, 2,814 in the main sample. Generic buyer labels that pool units across provinces, notably the Ministry of Health’s provincial directorates, were split by the province recorded for each tender (34 labels, 732 main-sample contracts). Each contract is assigned to one of 14 product markets from its title and each buyer to one of 11 sectors.

To validate the market classification, two AI coders (different language models), blind to each other and to the rules, labelled a random sample of 300 titles. They agreed with each other in 89% of cases (Cohen’s $\kappa = 0.87$). The rules agreed with their joint label in 79% of the 266 titles on which both coders agreed, with precision and recall of 1.00 and 0.99 for health information systems. These are AI rather than human labels; a human check remains desirable. Values are deflated to 2025 lira with the consumer price index. In the released data, winners that are natural persons or lack an explicit legal form are pseudonymized. Tables 1 and 2 describe the sample.

3.3 Estimated costs

For a blind sample of 150 repeat-eligible tenders, 75 won by incumbents and 75 by other firms, we retrieved the estimated cost (*yaklaşık maliyet*), contract value, and product codes (OKAS, the Turkish CPV) from EKAP’s tender-detail pages. An AI agent read these pages in the authors’ ordinary browser session at a slow pace; its input list did not indicate which tenders were won by incumbents. Collection stopped at 144 tenders when the site displayed a human-verification check, which we did not attempt to bypass. The EKAP contract values match our records within 1% in 99% of cases, and 143 tenders have both values in lira.

Table 2 Product markets (main sample). Eligible: repeat-eligible contracts. Value in bn 2025 TRY; procedure shares in % of contracts.

Product market	Contracts	Eligible	Firms	Value	21(b)	21(f)
Health information systems	2,189	1,252	205	19.48	34.7	9.4
ERP / management software	1,515	745	599	11.14	4.5	47.1
Custom software / web / mobile	1,117	488	734	7.04	2.9	43.2
Software licences	1,047	602	375	7.68	1.8	23.4
Network / data-centre infrastructure	748	328	390	7.43	5.9	25.0
Computers / peripherals	696	281	467	2.85	3.9	27.7
Maintenance / support services	626	375	297	8.23	3.5	33.2
GIS / city information	553	230	203	4.75	2.2	41.4
Cybersecurity	390	161	194	3.25	5.4	33.1
Physical security / surveillance	385	182	222	2.29	1.8	23.1
Other IT	277	112	215	1.82	5.4	27.8
Education technology	205	80	128	1.87	5.4	50.2
Smart city / traffic	169	58	106	3.44	4.1	34.3
Call centre / help desk	74	19	46	7.99	5.4	51.4
All	9,991	4,913	2,814	89.26	10.5	29.6

4 Methods

4.1 Concentration

For each product market we compute the supplier HHI ($H = \sum_i s_i^2$, 0–10,000) and the four-firm ratio from contract counts and real values. Because single contracts can dominate value shares, we also report value HHIs without the largest contract and with winsorized values, with bias-shifted bootstrap intervals. As reference points we use 1,000 and 1,800, the “moderately” and “highly” concentrated boundaries of the 1982 and 1992 US merger guidelines; the 2023 guidelines kept 1,800 ([U.S. Department of Justice and Federal Trade Commission, 1992, 2023](#)).

4.2 Incumbency, renewals, and null models

A contract awarded by buyer k in product market m on date t is *repeat-eligible* if k awarded another contract in m strictly before t , and *incumbent* if its winner had won from k in m strictly before t . The incumbency rate is the share of repeat-eligible contracts that are incumbent. A repeat-eligible contract is a *renewal* if the buyer awarded a contract in the same market 6–18 months earlier with a similar title (Jaccard similarity of at least 0.5 between normalized, stemmed title words); otherwise it is a *new need*. In the blind AI coding of 60 title pairs, 35–40 of the 42 pairs above the threshold were judged the same service, but so were 6–10 of the 18 pairs just below it. Some renewals are therefore classified as new needs, which biases the contrast between the groups towards zero.

To separate persistence from market structure, we permute winners across contracts within cells and recompute all statistics (1,000 permutations each). This keeps each buyer’s demand and each firm’s number of wins in every cell, and hence market size, specialization, and entry and exit, while randomizing only who wins from whom.

Table 3 lists the cells. N4 and N4b use each firm’s home province (the province of most of its wins before the tender), which partly reflects earlier wins from the same buyer and makes these nulls conservative. With 1,000 draws the smallest attainable p -value is 0.001, reported as $p < 0.001$.

4.3 Who wins, and why

The nulls randomize winners among all firms in a cell. A sharper benchmark compares the winner with plausible alternatives. For each repeat-eligible contract we take as alternatives all firms that won in the same market in the 24 months before the tender. We then estimate a conditional logit of who wins among them (McFadden, 1974, 1978). This describes the outcome of participation and award together, not the buyer’s choice alone, since in open tenders the lowest compliant bid wins. The covariates are:

- whether the firm supplied the buyer in the market before (incumbent);
- whether it supplied the buyer in another market;
- whether its home province is the buyer’s;
- its recent activity in the market, total experience, and experience with buyers of the same sector.

All covariates are measured before the tender, and standard errors are clustered by buyer. Tenders won by firms not active in the market in the previous 24 months (entrants, 39% of repeat-eligible contracts) cannot be represented in these choice sets; we report estimates with and without them.

To ask how much of the incumbent advantage is state dependence, we use a lead-lag placebo (Heckman & Borjas, 1980): an indicator for firms that will supply the buyer in the market only after the tender date. Under pure state dependence, future suppliers should be much less favoured than past ones; under persistent heterogeneity, the two should be similar. Because the design itself makes future suppliers look favoured, we benchmark the observed moments against two simulated worlds (100 replications each). In the state-dependence world, winners are redrawn in chronological order from the fitted model, with incumbency and all other covariates recomputed from the simulated history. In the heterogeneity world there is no incumbency effect, but a persistent random buyer–firm effect is calibrated to reproduce the observed share of incumbent winners. We compare three moments across the observed data and the two worlds: the past-supplier odds ratio, the future-supplier odds ratio, and the ratio between the two, separately for renewals and new needs. Intermediate worlds that mix both mechanisms are reported in the Supplementary Information.

4.4 Procedure, bids, and prices

We relate incumbency to the procedure with a contract-level logit with product-market, year, and buyer-sector fixed effects and two-way clustered errors (buyer and winner) (Cameron, Gelbach, & Miller, 2011), and describe its timing with an event study (Supplementary Information). Bid counts come from result announcements for a year-stratified random sample of 393 tenders (2012–2026, design-weighted) and a

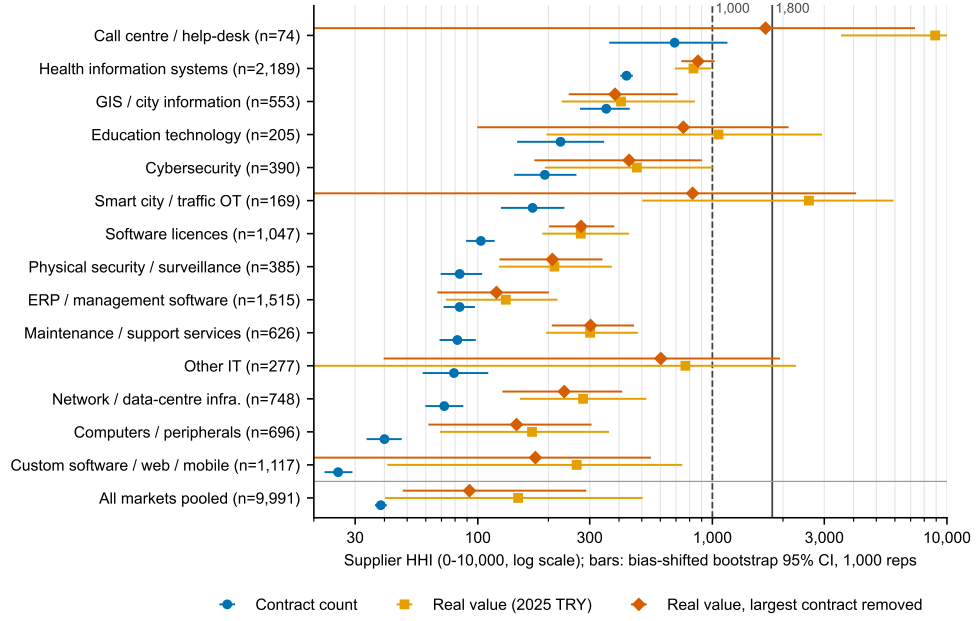


Fig. 1 Supplier concentration by product market: HHI by contract count, by real value, and by real value without the market’s largest contract, with bias-shifted bootstrap 95% intervals (1,000 resamples). Lines mark 1,000 and 1,800. Log scale.

larger non-random sample of 1,194 software tenders from 2021–2026. In the estimated-cost sample we compare the discount, $1 - \text{contract value}/\text{estimated cost}$, between incumbent and other winners with ordinary least squares and robust errors.

5 Results

5.1 Supply is not concentrated

Pooled over all markets, the supplier HHI is 39 by count and 149 by real value (92 without the largest contract). By count, every product market is below 1,000 (Fig. 1); the highest are call-centre services (690, 74 contracts) and health information systems (431, 95% CI 405–458). Value-based HHIs exceed 1,000 only in three small markets dominated by one or two contracts. Health information systems, with 2,189 contracts and 205 firms, is the one large market close to moderate concentration (value HHI 831, 692–985). The gap between pooled and market-level indices is mostly arithmetic dilution across markets (Supplementary Information). The point is that a concentration screen, however defined, raises no flag.

5.2 Buyers return to incumbents

Of the 4,913 repeat-eligible contracts, 45.2% went to an incumbent. The nulls in Table 3 predict between 6.2% and 26.4%, depending on how much of the market’s

Table 3 Incumbency rate against permutation nulls: winners are permuted within the listed cells (1,000 permutations each; all $p < 0.001$). Renewals: a similar title bought by the same buyer in the same market 6–18 months earlier. A dash: not computed for this null.

Cells: product market $\times$ year $\times$		All	Renewals	New needs
Observed		0.452	0.720	0.305
n		4,913	1,743	3,170
N1	–	0.062	0.092	0.045
N2	buyer sector	0.110	0.166	0.078
N3	buyer province	0.207	0.288	0.163
N4	winner’s home province	0.173	0.270	0.120
N4b	buyer province $\times$ winner’s home province	0.264	0.391	0.195
Observed / N4b		1.71	1.84	1.57

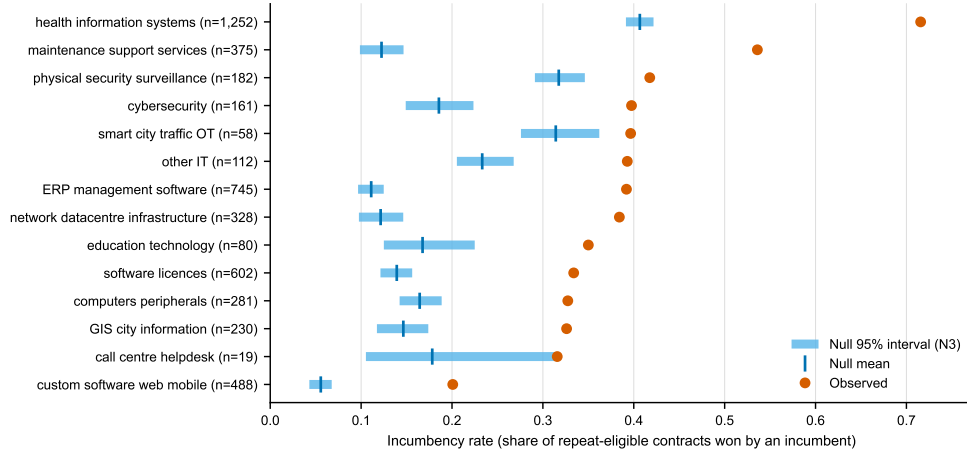


Fig. 2 Incumbency rate by product market: observed (dots) and permutation null N3 (mean and 95% interval; winners permuted within product market $\times$ year $\times$ buyer province). n is the number of repeat-eligible contracts.

structure they hold fixed. Observed incumbency exceeds every null ($p < 0.001$), by 1.7 times under the most conservative (N4b), and by about 920 contracts under that null. The excess holds in every year and every product market, including under the province null (Fig. 2). It is robust to recorded instead of split buyer labels and recorded instead of canonicalized firm names. It is also robust to excluding call-centre tenders, multi-lot tenders, natural-person winners, or named-product licence and maintenance renewals, and to including the ambiguous-scope contracts. Among the 540 buyers with at least five contracts, 51% have a main-supplier share above their own null after controlling the false discovery rate (Benjamini & Hochberg, 1995). The network of buyers and suppliers is only moderately modular once degrees are held fixed (Supplementary Information); what stands out is the persistence of individual ties, not segmentation.

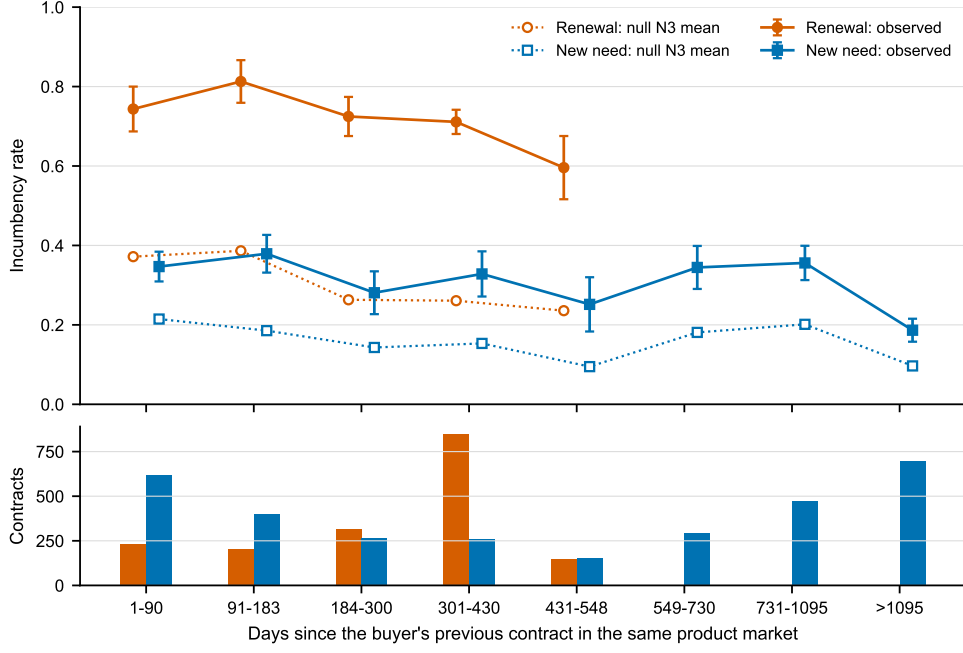


Fig. 3 Incumbency by days since the buyer's previous contract in the same product market, for renewals and new needs: observed (95% intervals) and null N3 mean. Lower panel: number of contracts.

5.3 Renewals and new needs

A third of repeat-eligible contracts (35.5%) renew a recurring service. Their incumbency rate is 72%, and they account for 57–64% of the excess, depending on the null. New needs go to incumbents 31% of the time, 1.6 times the most conservative null (Table 3), and even contracts with no similar earlier title go to incumbents 1.8 times as often as the province null predicts. The difference between renewals and new needs holds at every interval since the buyer's previous purchase (Fig. 3) and in each of the eight largest markets. In health information systems, 87% of renewals and 61% of new needs go to incumbents.

5.4 Who wins: state dependence or heterogeneity?

Table 4 reports the conditional logit. Among the 61% of repeat-eligible contracts won by a firm already active in the market, having supplied the buyer before raises a firm's odds of winning 42-fold (95% CI 36–49), holding constant its recent activity, experience, and location. Including entrants in the choice sets gives a lower bound of 22. The advantage is largest for renewals (141) and under Article 21(b) (96), and smaller for new needs (21) and open tenders (34).

The placebo and the simulations show what this advantage consists of (Fig. 4). Firms that will supply the buyer only in the future also have a large advantage: an

Table 4 Conditional logit of who wins among firms active in the market in the 24 months before each tender. Odds ratios with buyer-clustered 95% confidence intervals; 3,013 contracts won by established firms (909 buyers). The remaining 1,900 repeat-eligible contracts were won by entrants.

	Odds ratio [95% CI]
Incumbent: supplied this buyer in this market before	42.3 [36.4, 49.2]
Supplied this buyer in another market	2.17 [1.68, 2.81]
Home province (from prior wins) = buyer’s province	2.85 [2.44, 3.33]
Log wins in this market, previous 12 months	1.89 [1.71, 2.10]
Log total prior wins	0.91 [0.83, 1.01]
Log prior wins with buyers of the same sector	1.79 [1.60, 2.00]
Incumbent OR: renewals / new needs	140.7 / 21.0
Incumbent OR: Article 21(b) / open	95.8 / 33.7
Incumbent OR, entrants included (lower bound)	21.5 [18.5, 25.0]

odds ratio of 54 (42–70), close to the 64 of past suppliers in the same model. A world with pure state dependence cannot produce this. At the fitted incumbency effect it yields a future-supplier odds ratio of 20, and 22 or 24 when calibrated to the observed share of incumbent winners or when all covariates are recomputed from the simulated history. It also predicts past suppliers to be about three times as favoured as future ones, far above the observed 1.2 ($p < 0.001$). About a quarter of the future-supplier advantage reflects firms that are growing across all buyers. Controlling for their future wins with other buyers reduces it to 42. It survives counting only ties formed more than a year later (49), so it is not driven by lots or phases of the same project.

The subgroups differ. For new needs, future suppliers are at least as favoured as past ones (odds ratios 40 and 30), which is what heterogeneity predicts and the opposite of state dependence. For renewals, neither pure world fits: past suppliers are twice as favoured as future ones, far below the ratio of 8 under pure state dependence but above the 1.1 under pure heterogeneity. Renewals are therefore consistent with a mixture: strong persistent heterogeneity plus a state-dependence component that, in simulations of intermediate worlds, corresponds to an incumbency odds ratio of roughly 7 to 20. Recency does not discriminate between the explanations. The buyer’s most recent supplier is favoured over earlier ones (odds ratio 1.75), but heterogeneity alone produces the same pattern, because a displaced supplier has revealed a worse match.

5.5 Procedure

Incumbency is higher under Article 21(b) than under open tendering (72% against 44% of repeat-eligible contracts, a gap well above its null). Most excess incumbency nonetheless occurs in open tenders, which are far more numerous. Under Article 21(f), 35% of repeat-eligible contracts go to an incumbent, 2.6 times the province null. The 21(b)–open gap rose sharply around 2017–18, mainly among health buyers after hospital procurement was decentralized. The exact timing and cause are uncertain, and the Supplementary Information reports the event study and its caveats.

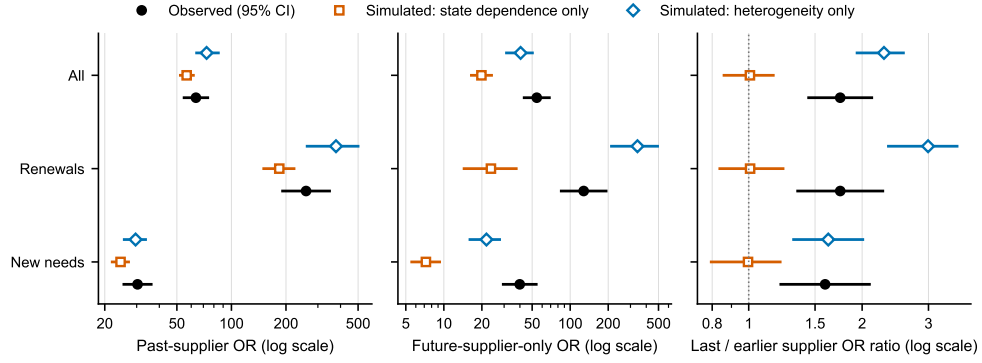


Fig. 4 Observed moments of the choice model (95% CI) and their values in simulated worlds with state dependence only and with persistent buyer–firm heterogeneity only (mean and 95% interval over 100 replications). Left: odds ratio for past suppliers; centre: for firms that supply the buyer only after the tender; right: ratio of the odds ratios for the most recent and earlier suppliers. Log scales.

5.6 Bids and prices

Incumbent wins coincide with weaker competition at the tender stage. In the random bid-count sample, tenders won by an incumbent drew a single bid in 71% of cases (95% CI 61–80%), against 32% (24–42%) for other winners (Fig. 5). With procedure, year, and renewal controls, the odds ratio is 3.6. In the larger software sample it is 2.7 for a single valid bid. These are associations: incumbents may win because others do not bid, or others may not bid because the incumbent is expected to win.

In the estimated-cost sample, contracts won by incumbents were awarded at a mean discount of 12.1% below the estimated cost, against 16.5% for other winners (medians 8.9% and 11.2%). The difference is 4–6 percentage points with or without controls for renewal, procedure, value, year, and market. It is not statistically significant (p between 0.11 and 0.22, $n = 143$). The point estimate is consistent with incumbents extracting part of the buyer’s switching cost, but a sample of this size cannot establish it.

6 Discussion

6.1 Interpretation

The market-level and buyer-level pictures differ. Many firms compete, and no product market is concentrated, yet buyers return to their suppliers far more often than market structure, timing, and geography imply. The decomposition qualifies the usual lock-in reading of this persistence. Most of it is persistent buyer–firm heterogeneity: firms that will supply a buyer later are almost as favoured as those that supplied it before, which switching costs cannot produce. Heterogeneity has benign and non-benign readings that our data cannot separate. It may reflect specialization, compatibility with installed systems, geography, or good past performance (Coviello et al., 2018; Decarolis et al., 2016). It may equally reflect relationships that favour certain suppliers or the anticipation of multi-stage projects, as the Turkish evidence on connections suggests

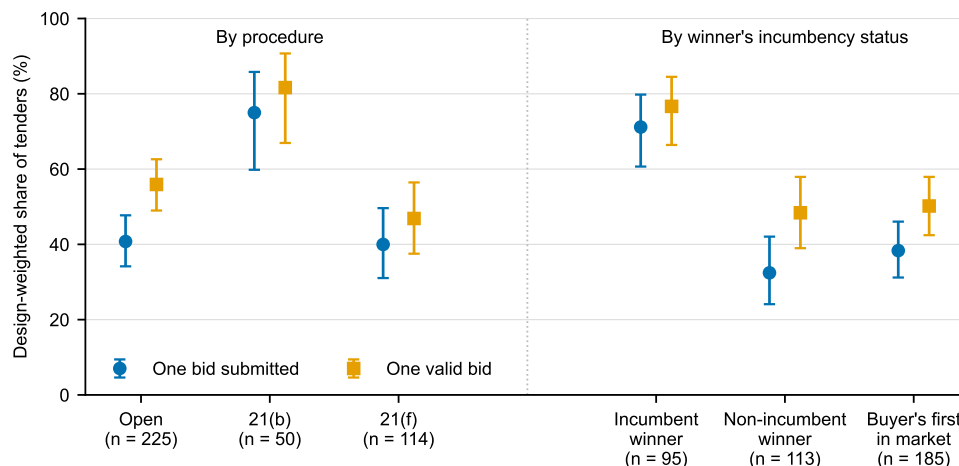


Fig. 5 Single-bid rates by procedure and by whether the winner was an incumbent (design-weighted, year-stratified sample of 393 tenders, 2012–2026; 95% intervals).

(Gürakar, 2016; Kutlay & Yıldırım, 2026). Switching costs are visible where theory predicts: in renewals of recurring services, where the incumbent advantage exceeds what heterogeneity alone implies. New needs show no sign of state dependence.

Incumbent wins attract single bids far more often. In a small sample, they also carry smaller discounts on the estimated cost. Both patterns fit weaker competition when incumbents are present. Both also fit purchases that only the incumbent can supply. Procedure is one channel of persistence, not its source: persistence is highest under the negotiated exception route, but most of it arises in open tenders.

6.2 Implications for monitoring

Indicators of buyer–supplier concentration already exist in the integrity literature (Fazekas & Kocsis, 2020; Fazekas et al., 2024). What our analysis adds is a benchmark and a decomposition. Buyer-level incumbency can be compared with permutation nulls that account for market size, timing, specialization, and geography. It can be separated into renewals, where switching costs are plausible, and new needs, where persistence reflects heterogeneity whose nature needs case-level review. Such benchmarked indicators identify candidates for scrutiny, not violations. Where switching costs are the concern, lowering them through interoperability and data portability or separate procurement of licences and maintenance is the natural remedy (European Commission, 2013). Where persistence reflects heterogeneity, the relevant questions concern how buyers use discretion.

6.3 Limitations

The evidence is observational. The decomposition rests on a placebo and on simulated benchmarks from one model class, not on a structural model with bidders. We observe winners, bid counts for samples, and estimated costs for 143 tenders, but

not losing bidders’ identities or quality, so welfare effects are not established. Scope, product markets, and renewals are classified from titles, and their validation uses AI coders; human coding remains desirable. Firm names were canonicalized without tax identifiers. Reverting to recorded firm names lowers observed incumbency (0.394), and reverting to recorded buyer labels raises it (0.462); the excess over the nulls persists in both cases. Multi-lot tenders appear with a single winner.

7 Conclusion

Türkiye’s public IT market is unconcentrated, yet its buyers keep returning to the same suppliers. Most of this persistence is stable heterogeneity between buyers and firms rather than lock-in. Switching costs appear in the renewal of recurring services, where the incumbent advantage exceeds what heterogeneity implies. Market concentration is therefore an incomplete guide to competition in public procurement. Buyer-level persistence, benchmarked against null models and decomposed into renewals and new needs, shows where to look and what kind of problem to look for.

Supplementary information. The Supplementary Information contains the scope audit and classification checks, concentration tables, all null-model results and robustness checks, the renewal classification, the choice model and simulation details, the procedure and timing analysis with the statutory texts, the bid and price samples, and the network analysis.

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Declarations

- **Funding.** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Competing interests.** The authors have no competing interests to declare.
- **Ethics approval.** Not applicable. The study uses public administrative records; names of natural persons are pseudonymized in the released data.
- **Data and code availability.** Code, derived data, and a data dictionary are available at <https://github.com/XXXX/TenderNet> and archived on Zenodo (<https://doi.org/10.5281/zenodo.XXXXXXX>). The underlying records are published by the Public Procurement Authority on EKAP (<https://ekapv2.kik.gov.tr>).
- **Use of AI tools.** Claude (Anthropic) was used to assist with analysis code, figure preparation, language editing, and data collection. For the validation of classifications, two instances based on different models independently labelled samples of tender titles (disclosed in Section 3). An AI agent also retrieved estimated costs from EKAP’s public pages through the authors’ own browser at a slow pace, without bypassing any verification. The authors designed the study, checked the code and results, and take full responsibility for the content.

- **Author contributions.** F.A.: conceptualization, methodology, data curation, software, formal analysis, visualization, writing (original draft, review and editing). A.Ş.: data curation, investigation, validation, visualization, writing (review and editing).

References

- Alsaç, U. (2017). EKAP: Turkey’s centralized e-procurement system. R.K. Shakya (Ed.), *Digital governance and e-government principles applied to public procurement* (pp. 126–150). IGI Global.
- Alyanak, S. (2022). 4734 sayılı kamu ihale kanununun 21(b) hükmünün Avrupa Birliği kamu alımları mevzuatı çerçevesinde analizi. *Hacettepe Hukuk Fakültesi Dergisi*, 12(2), 1128–1161, <https://doi.org/10.32957/hacettepehdf.1092034>
- Bajari, P., McMillan, R., Tadelis, S. (2009). Auctions versus negotiations in procurement: An empirical analysis. *Journal of Law, Economics, and Organization*, 25(2), 372–399, <https://doi.org/10.1093/jleo/ewn002>
- Baltrunaite, A., Giorgiantonio, C., Mocetti, S., Orlando, T. (2021). Discretion and supplier selection in public procurement. *The Journal of Law, Economics, and Organization*, 37(1), 134–166, <https://doi.org/10.1093/jleo/ewaa009> (Published online 15 May 2020)
- Bandiera, O., Prat, A., Valletti, T. (2009). Active and passive waste in government spending: Evidence from a policy experiment. *American Economic Review*, 99(4), 1278–1308, <https://doi.org/10.1257/aer.99.4.1278>
- Bařanek, B., & Titl, V. (2024). The cost of favoritism in public procurement. *The Journal of Law and Economics*, 67(2), 445–477, <https://doi.org/10.1086/727793>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300, <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Best, M.C., Hjort, J., Szakonyi, D. (2023). Individuals and organizations as sources of state effectiveness. *American Economic Review*, 113(8), 2121–2167, <https://doi.org/10.1257/aer.20191598>

- Board, S. (2011). Relational contracts and the value of loyalty. *American Economic Review*, 101(7), 3349–3367, <https://doi.org/10.1257/aer.101.7.3349>
- Borenstein, S., MacKie-Mason, J.K., Netz, J.S. (2000). Exercising market power in proprietary aftermarket. *Journal of Economics & Management Strategy*, 9(2), 157–188, <https://doi.org/10.1111/j.1430-9134.2000.00157.x>
- Buğra, A., & Savaşkan, O. (2014). *New capitalism in Turkey: The relationship between politics, religion and business*. Cheltenham: Edward Elgar.
- Butler, J.V., Carbone, E., Conzo, P., Spagnolo, G. (2020). Past performance and entry in procurement: An experimental investigation. *Journal of Economic Behavior & Organization*, 173, 179–195, <https://doi.org/10.1016/j.jebo.2020.02.019>
- Calzolari, G., & Spagnolo, G. (2009). *Relational contracts and competitive screening* (CEPR Discussion Paper No. 7434). London: Centre for Economic Policy Research. Retrieved from <https://cepr.org/publications/dp7434> (No journal version located)
- Cameron, A.C., Gelbach, J.B., Miller, D.L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2), 238–249, <https://doi.org/10.1198/jbes.2010.07136>
- Coviello, D., & Gagliarducci, S. (2017). Tenure in office and public procurement. *American Economic Journal: Economic Policy*, 9(3), 59–105, <https://doi.org/10.1257/pol.20150426>
- Coviello, D., Guglielmo, A., Spagnolo, G. (2018). The effect of discretion on procurement performance. *Management Science*, 64(2), 715–738, <https://doi.org/10.1287/mnsc.2016.2628>
- Decarolis, F., Fisman, R., Pinotti, P., Vannutelli, S. (2025). Rules, discretion, and corruption in procurement: Evidence from Italian government contracting. *Journal of Political Economy Microeconomics*, 3(2), 213–254, <https://doi.org/10.1086/732654> (Published version of NBER Working Paper 28209 (2020); cite the journal version.)
- Decarolis, F., Spagnolo, G., Pacini, R. (2016). *Past performance and procurement outcomes* (NBER Working Paper No. 22814). National Bureau of Economic Research.

- De Silva, D.G., Dunne, T., Kosmopoulou, G. (2003). An empirical analysis of entrant and incumbent bidding in road construction auctions. *The Journal of Industrial Economics*, 51(3), 295–316, <https://doi.org/10.1111/1467-6451.00202>
- Dubé, J.-P., Hitsch, G.J., Rossi, P.E. (2010). State dependence and alternative explanations for consumer inertia. *The RAND Journal of Economics*, 41(3), 417–445, <https://doi.org/10.1111/j.1756-2171.2010.00106.x>
- European Commission (2013). *Against lock-in: Building open ICT systems by making better use of standards in public procurement*. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, COM(2013) 455 final. Retrieved from <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52013DC0455> (Accompanied by SWD(2013) 224 final, “Guide for the procurement of standards-based ICT – Elements of good practice”)
- Farrell, J., & Klemperer, P. (2007). Coordination and lock-in: Competition with switching costs and network effects. M. Armstrong & R. Porter (Eds.), *Handbook of industrial organization* (Vol. 3, pp. 1967–2072). Amsterdam: Elsevier.
- Farrell, J., & Shapiro, C. (1988). Dynamic competition with switching costs. *The RAND Journal of Economics*, 19(1), 123–137, <https://doi.org/10.2307/2555402>
- Fazekas, M., & Kocsis, G. (2020). Uncovering high-level corruption: Cross-national objective corruption risk indicators using public procurement data. *British Journal of Political Science*, 50(1), 155–164, <https://doi.org/10.1017/S0007123417000461>
- Fazekas, M., Tóth, B., Abdou, A., Al-Shaibani, A. (2024). Global contract-level public procurement dataset. *Data in Brief*, 54, 110412, <https://doi.org/10.1016/j.dib.2024.110412> (Defines the buyer spending concentration indicator (corr_spending.concentration): supplier’s share of a buyer’s total awarded amount)
- Fazekas, M., Tóth, I.J., King, L.P. (2016). An objective corruption risk index using public procurement data. *European Journal on Criminal Policy and Research*, 22(3), 369–397, <https://doi.org/10.1007/s10610-016-9308-z>

- Fazekas, M., & Wachs, J. (2020). Corruption and the network structure of public contracting markets across government change. *Politics and Governance*, 8(2), 153–166, <https://doi.org/10.17645/pag.v8i2.2707>
- Fountoukidis, I., Daffi, E., Antoniou, I.E., Varsakelis, N. (2026). *A multi-layered diagnostic framework for public procurement markets: Linking market structure, relational patterns, and institutional dynamics*. Zenodo preprint. Zenodo. (Preprint)
- Fountoukidis, I.G., Antoniou, I.E., Varsakelis, N.C. (2023). Competitive conditions in the public procurement markets: an investigation with network analysis. *Journal of Industrial and Business Economics*, 50(2), 347–368, <https://doi.org/10.1007/s40812-022-00251-z>
- Greenstein, S.M. (1993). Did installed base give an incumbent any (measurable) advantages in federal computer procurement? *The RAND Journal of Economics*, 24(1), 19–39, <https://doi.org/10.2307/2555951>
- Gürakar, E.Ç. (2016). *Politics of favoritism in public procurement in Turkey: Reconfigurations of dependency networks in the AKP era*. New York: Palgrave Macmillan.
- Gürakar, E.Ç., & Bircan, T. (2019). Political connections and public procurement in Turkey: Evidence from construction work contracts. I. Diwan, A. Malik, & I. Atiyas (Eds.), *Crony capitalism in the Middle East: Business and politics from liberalization to the Arab spring* (pp. 236–262). Oxford: Oxford University Press.
- Heckman, J.J. (1981). Heterogeneity and state dependence. S. Rosen (Ed.), *Studies in labor markets* (pp. 91–140). Chicago: University of Chicago Press. Retrieved from <https://www.nber.org/books-and-chapters/studies-labor-markets/heterogeneity-and-state-dependence>
- Heckman, J.J., & Borjas, G.J. (1980). Does unemployment cause future unemployment? definitions, questions and answers from a continuous time model of heterogeneity and state dependence. *Economica*, 47(187), 247–283, <https://doi.org/10.2307/2553150>
- Hirschman, A.O. (1964). The paternity of an index. *American Economic Review*, 54(5), 761–762,
- Hyttinen, A., Lundberg, S., Toivanen, O. (2018). Design of public procurement auctions: Evidence from cleaning contracts. *RAND Journal of Economics*, 49(2),

398–426, <https://doi.org/10.1111/1756-2171.12232>

- Jofre-Bonet, M., & Pesendorfer, M. (2003). Estimation of a dynamic auction game. *Econometrica*, 71(5), 1443–1489, <https://doi.org/10.1111/1468-0262.00455>
- Kang, K., & Miller, R.A. (2022). Winning by default: Why is there so little competition in government procurement? *The Review of Economic Studies*, 89(3), 1495–1556, <https://doi.org/10.1093/restud/rdab051>
- Keane, M.P. (1997). Modeling heterogeneity and state dependence in consumer choice behavior. *Journal of Business & Economic Statistics*, 15(3), 310–327, <https://doi.org/10.1080/07350015.1997.10524709>
- Klemperer, P. (1987). Markets with consumer switching costs. *The Quarterly Journal of Economics*, 102(2), 375–394, <https://doi.org/10.2307/1885068>
- Kutlay, M., & Yıldırım, K. (2026). Authoritarian populist turn and market capture in Turkey: The political economy of public procurement. *Competition & Change*, 30(3-4), 763–782, <https://doi.org/10.1177/10245294241235390> (First published online 27 May 2024)
- Li, T., & Zheng, X. (2009). Entry and competition effects in first-price auctions: Theory and evidence from procurement auctions. *The Review of Economic Studies*, 76(4), 1397–1429, <https://doi.org/10.1111/j.1467-937X.2009.00558.x>
- McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. P. Zarembka (Ed.), *Frontiers in econometrics* (pp. 105–142). New York: Academic Press.
- McFadden, D. (1978). Modelling the choice of residential location. A. Karlqvist, L. Lundqvist, F. Snickars, & J.W. Weibull (Eds.), *Spatial interaction theory and planning models* (pp. 75–96). Amsterdam: North-Holland.
- Palguta, J., & Pertold, F. (2017). Manipulation of procurement contracts: Evidence from the introduction of discretionary thresholds. *American Economic Journal: Economic Policy*, 9(2), 293–315, <https://doi.org/10.1257/pol.20150511>
- Pliatsidis, A.C. (2024). Analyzing concentration in the Greek public procurement market: a network theory approach. *Journal of Industrial and Business Economics*,

51(2), 431–480, <https://doi.org/10.1007/s40812-023-00291-z>

- Popa, M. (2019). Uncovering the structure of public procurement transactions. *Business and Politics*, 21(3), 351–384, <https://doi.org/10.1017/bap.2019.1>
- Reeves-Latour, M., & Morselli, C. (2017). Bid-rigging networks and state-corporate crime in the construction industry. *Social Networks*, 51, 158–170, <https://doi.org/10.1016/j.socnet.2016.10.003>
- Rhoades, S.A. (1993). The herfindahl–hirschman index. *Federal Reserve Bulletin*, 79, 188–189,
- Saini, V. (2012). Endogenous asymmetry in a dynamic procurement auction. *RAND Journal of Economics*, 43(4), 726–760, <https://doi.org/10.1111/1756-2171.12006>
- Shapiro, C. (1995). Aftermarkets and consumer welfare: Making sense of Kodak. *Antitrust Law Journal*, 63(2), 483–511, Retrieved from <https://www.jstor.org/stable/40843290> (No DOI registered; JSTOR stable URL)
- Shapiro, C., & Varian, H.R. (1999). *Information rules: A strategic guide to the network economy*. Boston, MA: Harvard Business School Press.
- Shcherbakov, O. (2016). Measuring consumer switching costs in the television industry. *The RAND Journal of Economics*, 47(2), 366–393, <https://doi.org/10.1111/1756-2171.12131>
- Szucs, F. (2024). Discretion and favoritism in public procurement. *Journal of the European Economic Association*, 22(1), 117–160, <https://doi.org/10.1093/jeea/jvad017> (Published online 7 March 2023)
- U.S. Department of Justice and Federal Trade Commission (1992). *Horizontal merger guidelines*. <https://www.justice.gov/archives/atr/1992-merger-guidelines>. (Section 1.51: HHI below 1,000 unconcentrated; 1,000–1,800 moderately concentrated; above 1,800 highly concentrated. Revised 1997 (Section 4) with thresholds unchanged)

- U.S. Department of Justice and Federal Trade Commission (2023, December). *Merger guidelines*. https://www.ftc.gov/system/files/ftc_gov/pdf/2023_merger_guidelines_final_12.18.2023.pdf. (Issued December 18, 2023)
- Wachs, J., Fazekas, M., Kertész, J. (2021). Corruption risk in contracting markets: a network science perspective. *International Journal of Data Science and Analytics*, 12, 45–60,
- Wachs, J., & Kertész, J. (2019). A network approach to cartel detection in public auction markets. *Scientific Reports*, 9, 10818,